

Twin-head spread + repowered skill null

OOS 2019-10-19 → 2026-08-13 · n=2491 days · CPU only · COMBO untouched (v2.0-combo-final)

BACKTEST ONLY. Cleaned+floored 2.b data reused as-is. Stage T frozen. Context excluded. Zero GPU. A verdict is not overridden by any single cycle. Operative floor is BTC.

SPREAD SELECTION SKILL FALSE AUC failed the 0.52 bar	MODEL-V3 VIABLE YES rel Sharpe > 0, total ≥ BTC, MaxDD ≤ BTC
MODEL-V3 PRODUCT-GRADE YES rel Sharpe ≥ 0.30 and avg alt ≥ 5%	
REL SHARPE 0.916	BOOK VS BTC 986.0% / 714.0%
MAXDD VS BTC -75.9% / -76.6%	MEDIAN Θ 0.2
% TIME IN BTC 94.6%	AVG ALT / #NAMES 5.4% · 1.35
GATE ON 19.1%	FORCED EXITS 0

Pre-registered criteria (verbatim, before results)

SPREAD has SELECTION SKILL if, at h=14 or h=30: mean per-date RankIC(spread) ≥ +0.01 AND mean per-date AUC ≥ 0.52 AND the §2 gate passes for the spread metric. MODEL-V3 is VIABLE if on the full OOS window at median θ: (a) relative-line Sharpe > 0; (b) total ≥ BTC B&H; (c) MaxDD ≤ BTC B&H. MODEL-V3 is PRODUCT-GRADE if additionally relative-line Sharpe ≥ 0.30 AND average alt allocation ≥ 5% (non-degenerate book). Per-cycle honesty table mandatory; no single cycle overrides. Mechanical, no post-hoc adjustment.

Repowered skill null

Bias: every fold's null mean must satisfy the E.1b centering bound (AUC around 0.5, RankIC around 0). Skill passes if, for the judged signal, ≥ 5 of 6 folds exceed their null 95th percentile OR the Stouffer-combined z across the 6 folds is ≥ 3.0 . Symmetric: failure = PARKED, no override, no retest with different folds.

Death-in-position convention

A held coin whose data ends is force-exited at its last available close (no better information assumed). The count and PnL impact of such forced exits is reported in every backtest of this project.

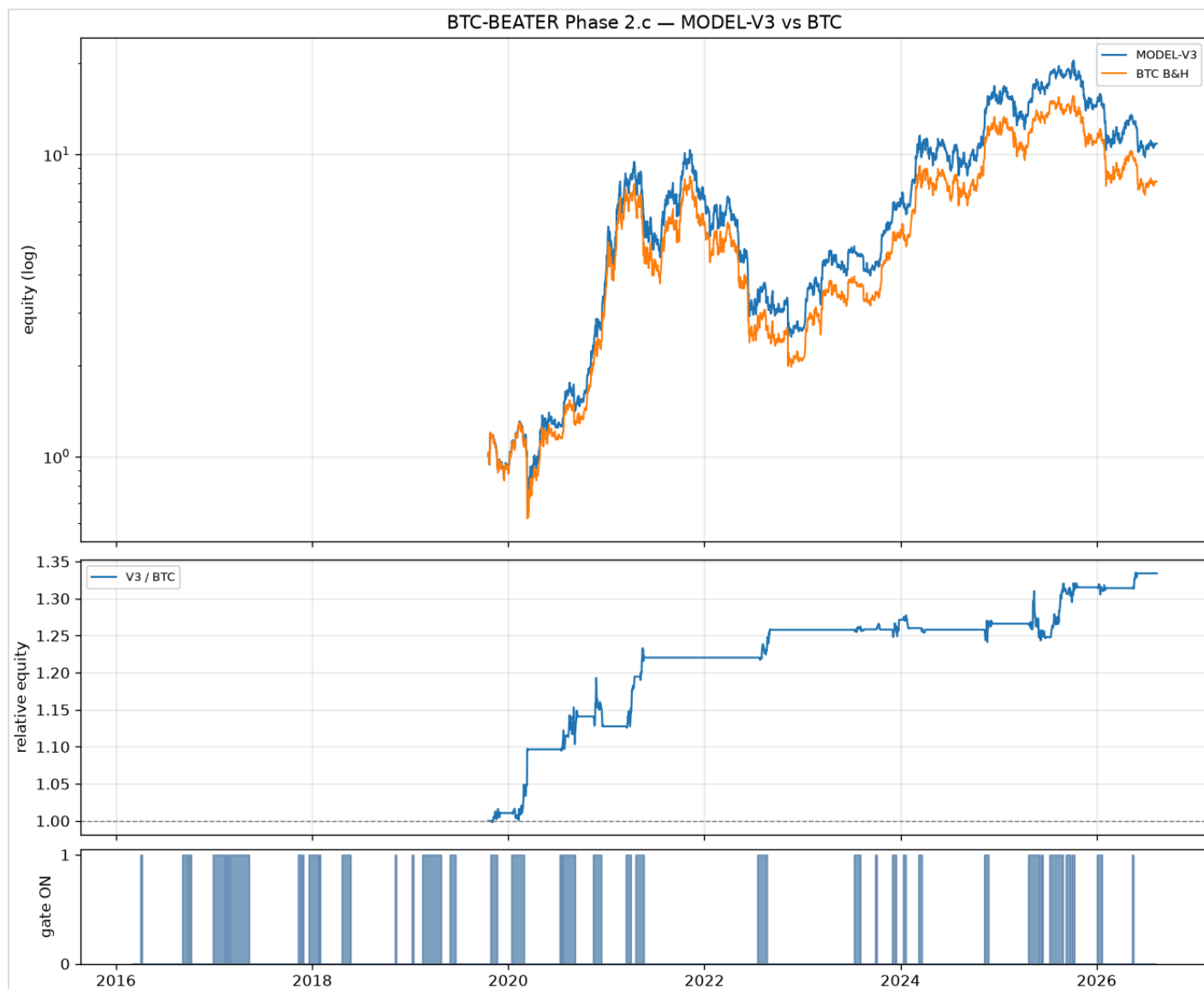
Mechanical verdicts

SPREAD has SELECTION SKILL: False (h=14 RankIC=0.1622 AUC=0.4884; h=30 RankIC=0.1911 AUC=0.5116; §2 spread RankIC GREEN 6/6 Stouffer $z=11.041$).

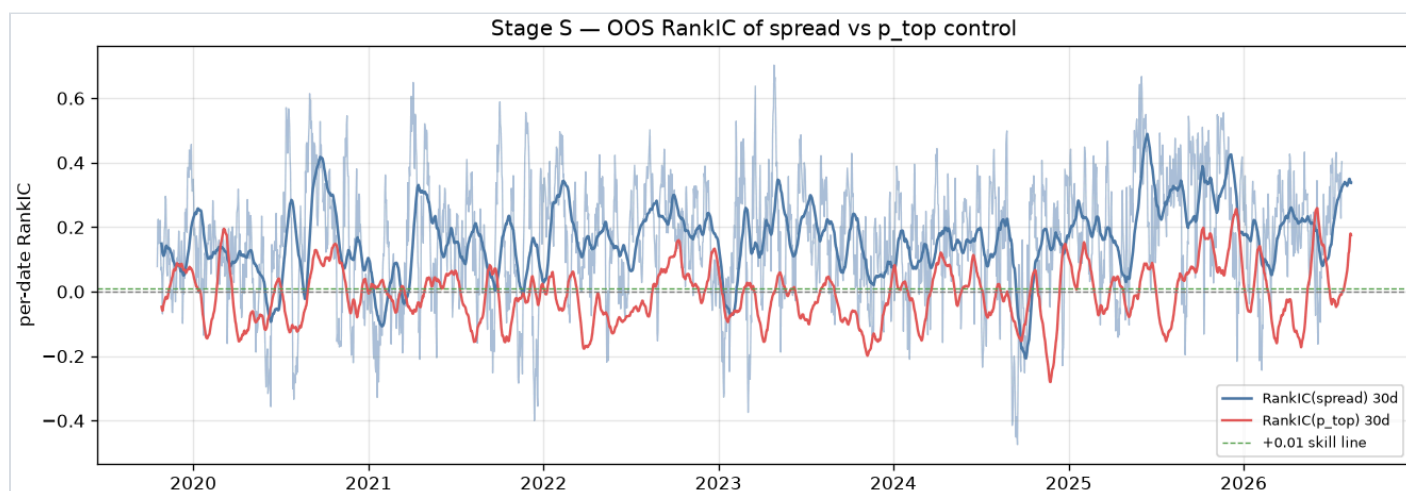
(a) rel Sharpe $0.916 > 0 \rightarrow \text{True}$. (b) book 986.0% vs BTC 714.0% $\rightarrow \text{True}$. (c) MaxDD -75.9% vs BTC -76.6% $\rightarrow \text{True}$. Product-grade need $\text{rel} \geq 0.300$ and $\text{alt} \geq 5.0\%$; avg alt 5.4%.

uncertainty $\leftrightarrow$ yz_vol_30 mean per-date RankIC = 0.5498 (lottery diagnostic; not used in trading). Naive v4 (record only): rel Sharpe=-0.670, live_benchmark=False.

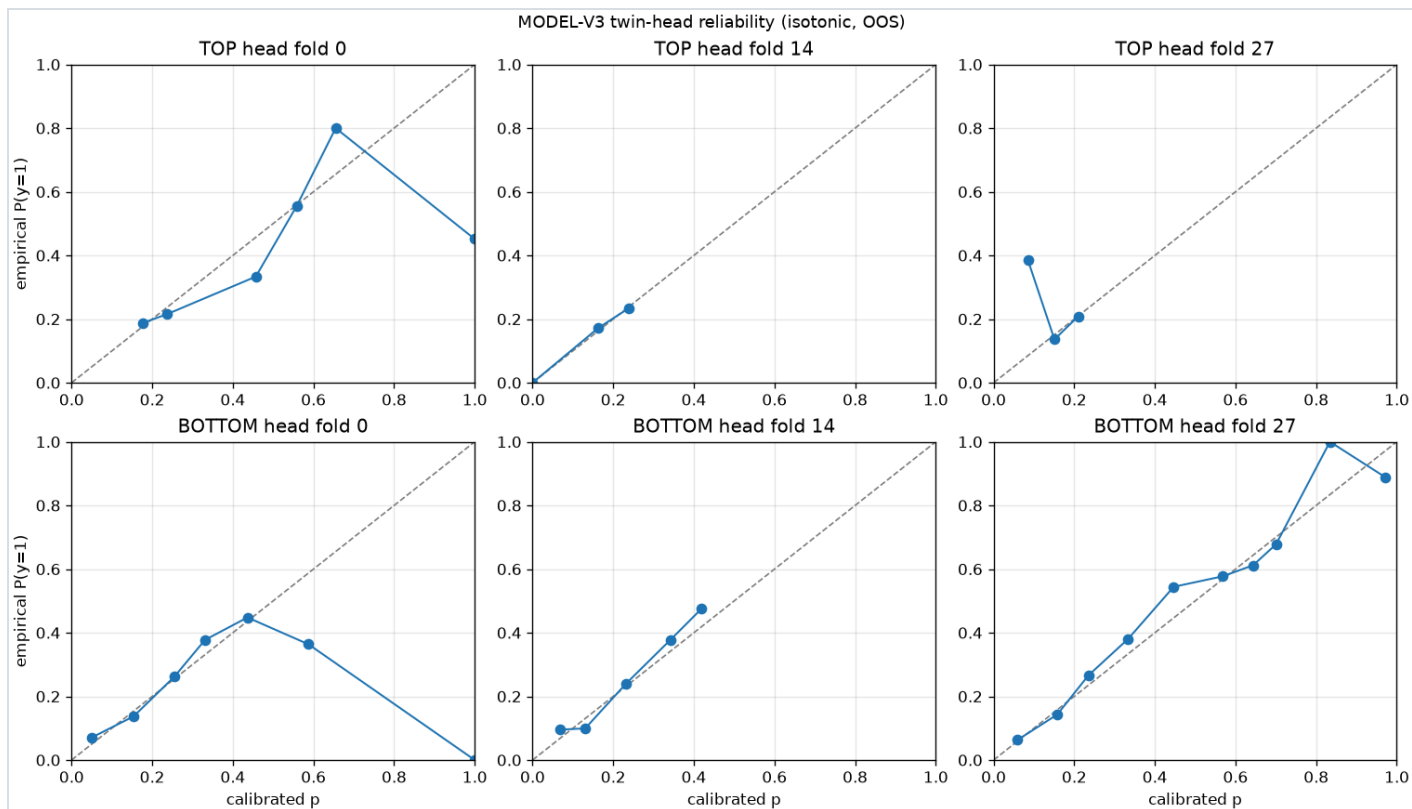
Charts



MODEL-V3 equity vs BTC (log) · relative line · gate ON



OOS RankIC of spread vs p_top control



Twin-head reliability (isotonic, OOS)

§2 null — p_{top} per-date AUC (h=14)

Bias pass=False; skill CONTAMINATED; 5/6 exceed p95; Stouffer z=7.190. Judged signal is spread RankIC, not this control.

fold	n	null mean	SD	95th	real	bias_ok	exceeds p95
0	25	0.5024	0.0212	0.5329	0.5349	True	True
5	25	0.5007	0.0285	0.5452	0.5556	True	True
9	25	0.5002	0.0220	0.5355	0.5334	True	False
15	25	0.5030	0.0211	0.5406	0.5654	True	True
21	25	0.4921	0.0299	0.5294	0.6230	True	True
24	25	0.4815	0.0409	0.5315	0.6986	False	True

§2 null — spread per-date RankIC (h=14, judged signal)

Bias pass=True; skill GREEN; 6/6 exceed p95; Stouffer z=11.041. Failure = PARKED, no override, no retest with different folds.

fold	n	null mean	SD	95th	real	bias_ok	exceeds p95
0	25	-0.0110	0.0295	0.0311	0.1314	True	True
5	25	0.0089	0.0349	0.0600	0.0831	True	True
9	25	0.0027	0.0382	0.0619	0.2540	True	True

fold	n	null mean	SD	95th	real	bias_ok	exceeds p95
15	25	0.0068	0.0367	0.0658	0.1827	True	True
21	25	-0.0047	0.0719	0.0846	0.2522	True	True
24	25	0.0035	0.0652	0.0804	0.3384	True	True

Per-fold selection metrics (floored PIT top-100)

h	fold	RankIC spread	RankIC p_top	AUC spread vs top-q	AUC p_top
14	0	0.1479	0.0319	0.4730	0.5348
14	1	0.1074	-0.0292	0.4819	0.5466
14	2	0.0555	-0.0658	0.4336	0.5265
14	3	0.2504	0.0372	0.5054	0.5379
14	4	0.0897	0.0085	0.4392	0.5477
14	5	0.0805	-0.0173	0.4688	0.5552
14	6	0.1679	0.0332	0.5337	0.5441
14	7	0.1419	-0.0739	0.4603	0.5593
14	8	0.0913	-0.0013	0.4697	0.5710
14	9	0.2240	-0.0568	0.5357	0.5312
14	10	0.1321	-0.0749	0.5233	0.5139
14	11	0.2508	0.0390	0.5200	0.5651
14	12	0.1723	0.0392	0.4911	0.5703
14	13	0.1015	-0.0531	0.4578	0.5168
14	14	0.2214	-0.0278	0.5393	0.5687
14	15	0.1718	-0.0313	0.4530	0.5635
14	16	0.0972	-0.0862	0.4376	0.5191
14	17	0.1153	-0.0207	0.4795	0.5446
14	18	0.1634	0.0503	0.5111	0.6156
14	19	0.0728	-0.0759	0.4394	0.5202
14	20	0.1115	-0.0181	0.4750	0.5513
14	21	0.2006	0.0061	0.4827	0.6067
14	22	0.2450	0.0253	0.5471	0.6126
14	23	0.2608	-0.0077	0.5412	0.5865
14	24	0.3237	0.1567	0.5582	0.6930
14	25	0.1441	-0.0348	0.4445	0.5675
14	26	0.1892	0.0190	0.4739	0.5506
14	27	0.2811	0.0202	0.5027	0.5396

h	fold	RankIC spread	RankIC p_top	AUC spread vs top-q	AUC p_top
30	0	0.1976	0.0178	0.5094	0.5242
30	1	0.0775	-0.0004	0.4951	0.5313
30	2	-0.0384	-0.0805	0.3983	0.4712
30	3	0.3245	0.1832	0.5560	0.5621
30	4	0.0735	-0.0393	0.4748	0.5055
30	5	0.1839	0.0747	0.5382	0.5568
30	6	0.1872	0.0594	0.5328	0.5556
30	7	0.1335	0.0446	0.4815	0.5347
30	8	0.1613	0.1414	0.4606	0.5606
30	9	0.2195	0.0315	0.5577	0.5449
30	10	0.1975	-0.0513	0.5532	0.5061
30	11	0.2954	0.1930	0.5451	0.6346
30	12	0.1955	0.0819	0.5345	0.5746
30	13	0.2574	0.1193	0.5575	0.5730
30	14	0.2343	0.0083	0.5536	0.5490
30	15	0.1492	0.0329	0.4524	0.5174
30	16	0.0892	0.0676	0.4578	0.5153
30	17	0.1707	-0.0097	0.5224	0.5178
30	18	0.1807	0.0512	0.5223	0.5821
30	19	-0.0302	-0.1248	0.3522	0.4015
30	20	0.2101	-0.0364	0.5602	0.5348
30	21	0.2227	0.0271	0.4693	0.5856
30	22	0.3236	0.0236	0.5805	0.5862
30	23	0.3716	0.1559	0.6294	0.6460
30	24	0.3044	0.2101	0.5260	0.6785
30	25	0.1960	0.0610	0.4834	0.5631
30	26	0.2506	0.0993	0.5099	0.5353
30	27	0.3632	-0.0387	0.5418	0.3871

Aggregate (last-fold-wins OOS): h=14 RankIC(spread)=0.1622 RankIC(p_top)=-0.0083 AUC(spread)=0.4884; h=30 RankIC(spread)=0.1911 RankIC(p_top)=0.0491 AUC(spread)=0.5116.

MODEL-V3 book vs BTC (same OOS window)

book	total	CAGR	USD Sharpe	rel Sharpe	MaxDD	avg #names	% BTC	gate ON	ann TO	forced
MODEL-V3 h=14 θ=0.2	986.0%	41.8%	0.889	0.916	-75.9%	1.35	94.6%	19.1%	1.92	0
BTC B&H	714.0%	36.0%	0.818	0.000	-76.6%	0.00	100.0%	0.0%	0.00	0

θ grid (h=14, median convention)

θ	rel Sharpe	total	MaxDD	% BTC	avg #names	gate ON
0.1	0.993	1319.2%	-75.0%	90.5%	4.72	19.1%
0.15	0.884	1137.7%	-75.3%	91.3%	3.09	19.1%
0.2 ← median	0.916	986.0%	-75.9%	94.6%	1.35	19.1%

Per-cycle honesty (headline h=14)

cycle	n	book tot	BTC tot	USD Sharpe	rel Sharpe	MaxDD	% BTC
2019-20	440	310.1%	263.7%	2.059	1.210	-47.6%	92.0%
2021	365	72.8%	59.7%	1.081	2.167	-51.7%	97.7%
2022	365	-63.2%	-64.3%	-1.251	1.627	-65.9%	98.3%
2023-24	731	468.3%	464.6%	2.031	0.135	-26.3%	96.6%
2025-26	590	-26.8%	-30.5%	-0.228	0.733	-52.1%	89.7%

Feature importances (mean gain, h=14)

Head TOP

rank	feature	mean gain
1	dist_high_90	8986.71
2	range_pos_28	7343.42
3	dist_ath	7312.88
4	log_age	7149.44
5	log_mcap	6934.78
6	beta_btc_60	5642.77
7	yz_vol_60	4722.74
8	idio_vol_60	4423.15
9	skew_60	4187.79
10	amihud_14	4026.91

Head BOTTOM

rank	feature	mean gain
1	idio_vol_60	62190.15
2	pk_vol_14	24091.91
3	log_age	20364.04
4	yz_vol_60	15220.73
5	corr_btc_28	14533.66
6	dist_low_90	10554.82
7	close_sma20	10466.89
8	yz_vol_30	9656.13
9	beta_btc_60	9626.70
10	dist_ath	8648.92

rank	feature	mean gain
11	corr_btc_28	3999.79
12	dist_low_90	3484.20
13	sma20_sma50	3386.45
14	yz_vol_30	3374.28
15	mcap_rank	3353.54

rank	feature	mean gain
11	log_mcap	8425.78
12	yz_vol_14	7910.97
13	turnover	7175.00
14	d_rank_90	6983.94
15	amihud_14	5930.66

Elapsed s=7873.0. GPU=False. n_features=33. COMBO untouched (v2.0-combo-final). Source: reports/btcb_phase2c_report.json — numbers are not recomputed.